

Technology Plus

Unit 6: Database

5.2.1 – Database Fundamentals

Lesson Overview:

In this lesson, students will explore the concepts of a database and the benefits of using a database when storing and managing data on a large scale.

Students will:

- Understand database uses.
- Understand the difference between a flat file and a database.
- Recognize how storage and access methods impact data availability in different scenarios.

Guiding Question: What are the uses of a database, when should it be used instead of a flat file, and how can someone choose the right storage and access methods when handling data?

Suggested Grade Levels: 8-12

Technology Needed: Yes

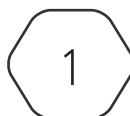
Approximate Time Required: 75-120 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 5.2 – Explain database concepts and the purpose of a database.
 - Database uses
 - Create
 - Import/input
 - Query
 - Reports
 - Flat file vs. database
 - Multiple concurrent users
 - Scalability
 - Speed
 - Variety of data
 - Database records
 - Storage
 - Data persistence
 - Data availability
 - Cloud vs. local
 - Online vs. offline

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Lesson Background

Background Information:

This is students' first exposure to databases, and the shift from spreadsheets or flat files to relational systems might require some adjustment. They'll recognize how data is structured and stored, but terms like *schema*, *primary key*, and *constraints* will be new and potentially abstract without strong examples.

Use this lesson to anchor their understanding of why databases matter—not just as a concept, but as a tool for solving real-world data problems. Comparisons between flat files and databases, local vs. cloud storage, and online vs. offline access will give students a framework for evaluating data systems. Don't expect mastery of SQL yet—this lesson is about making sense of structure and purpose before the syntax shows up.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 5.2.1 – Database Fundamentals • Activity Slides 5.2.1 – Storage and Access • Lab Slides 5.2.1 – SQL Databases
Materials per Student	• Student Notes 5.2.1 – Database Fundamentals • Student Handout 5.2.1 – Database Fundamentals • Assessment 5.2.1 – Database Fundamentals (optional)

Cyber Terms:

Term	Definition
data	distinct pieces of information, usually formatted in a special way
information	data that has been processed, organized, or structured in a way that makes it meaningful
critical data	essential information an organization needs to operate, meet legal requirements, or protect its customers
non-critical data	information that supports an organization's work but isn't essential for day-to-day operations or legal requirements

Lesson Background

Cyber Terms:

Term	Definition
database	a highly organized, multi-table digital storage system designed to hold information on a computer
database management system (DBMS)	software that enables users to define, create, maintain, and control access to databases
schema	the rules and structure of a database
data input	manually adding data directly into the database
data import	bringing data into the database from external sources
query	retrieving specific data from a database through asking a question
flat file	a spreadsheet used to manage information with a simple two-dimensional structure of rows and columns
records	a collection of fields that contain data about a given entity, also known as a row
data value	an individual piece of data
data persistence	the ability of a system to maintain data across sessions and reboots
data availability	the ability of a system to ensure that data is accessible and functional when needed

5.2.1 – Database Fundamentals

Guiding Question: What are the uses of a database, when should it be used instead of a flat file, and how can someone choose the right storage and access methods when handling data?

Lesson Launch: (3-5 minutes)

Is a database necessary?

- Prompt students to think about how often they use their phone and what they use it for.
- Ask students to reflect on their answers to the following questions:
 - What if you had to rely on books to gather information?
 - What if you had to memorize all your friends' phone numbers?
 - What if you could only communicate with people by writing letters?
 - What if you could only use photo albums to store any pictures you take?
- Tell students that smartphones are like personal databases; they make life convenient for the users. Databases are just as important to businesses.

What is a database?

- Remind students they have learned about the value of data and information, and how it can be utilized by businesses to inform their decisions.
- Introduce a **database** and a **database management system (DBMS)** and how they are used.

- **From Data to Database (20-35 minutes)**

- Distribute Student Handout 5.2.1 - Database Fundamentals and encourage students to fill in the answers during the lesson.

Database Uses

- Introduce the database creation process, discuss the images from the Linux terminal on slides 5 and 6.
 - Slide 5, In our example we chose SQLite3 as our DBMS.
 - Slide 5, **schema** shows a table named 'employees', each field (column) in the table, and the data type.
 - Slide 6, shows a table created using SQLite3 inside the Linux terminal.
- Mention students will learn more about schema and defining constraints in the next lesson (5.3.1 – Database Structures).
- Discuss **data input** and **data import**, explain the differences between each, discuss the images, and highlight why import is the best method for putting data in the system.

- Data input is when data is manually added to a database, it leaves room for human error and can be time consuming. It is never the best choice. The first image shows a query ran using SQL commands to input data into a database.
- Data import brings data into a database from external sources (flat files), it allows large amounts of information to be brought in quickly and accurately. It is the best choice for putting data into a database. The second image shows the commands for importing a comma-separated values (csv) file into a database using SQLite3.
- Discuss the importance of a **query**, they allow users to retrieve data from a database to easily find the information they are looking for. Use the example below to demonstrate how impactful queries are in a database. Explain the queries and relating images on slide 9.
 - Real-life example: Searching the internet is often faster than reading through a book to find a specific answer. For example, a student might run a web search to get the information they need in seconds.
 - Slide 9, image one shows a data retrieval query using the SELECT statement to retrieve all records in the 'employees' table. Note: using the asterisk (*) symbol in a query retrieves all information in the specified table, and a semicolon (;) must be at the end of each query for it to run properly.
 - Image two shows a filter query using the WHERE clause for the name of those in the 'employees' table who have an IT Support position.
 - Image three shows a sorting query using the ORDER BY clause to sort employees by their position.
- Briefly discuss reports and how they make data easier to read.
 - Reports were previously discussed in lesson 5.1.2.

Flat File vs. Database

- Introduce a **flat file**, highlight the differences between a flat file and a database.
 - On slide 11, students may not see much difference between the flat file and database screenshots. They should know the real difference is in how the two systems operate and handle data. Discuss considerations for choosing between the two.
 - While databases are optimal when compared to flat files, a flat file is useful when working with a small group of people and small to medium data sets. Databases should always be chosen when managing large amounts of data. Depending on software and experience, users can have a learning curve when choosing a database over a flat file.

Database Records

- Introduce a database **record**, the structure, characteristics, and example SQL operations.
 - Slide 13, shows a table with the fields: Name, Position, and Salary; and records for three employees. One record consists of every field for an individual employee, i.e. Debbie Butler, IT Support, 75366, is one record.
 - On slide 14, highlight the difference between a record and data values. A data value will be every entry put into a field, i.e. every name under the field 'Name' is a data value.
 - Mention students will learn more about primary key in the next lesson (5.3.1 – Database Structures).
 - Slide 16, shows SQL operations that can be run as queries using SQLite3 in a terminal .
 - Image one shows an insertion query into a table named 'employees' with data values for the employee's name, position, and salary.
 - Image two shows an update query in the 'employees' table for an existing employee. SET tells the database what data values to update and WHERE tells the database which employee is receiving the update.
 - Image three shows a deletion query in the 'employees' table for a specific employee.

Storage and Persistence

- Students should be familiar with storage from previous units in the Technology Plus Course. This section of the lesson briefly discusses storage as it relates to data availability, management, and persistence. Highlight the storage types that are impactful when using a database.
 - Files and Directories – files can represent a database or components of a database.
 - Tables and Indexes – a database is organized into tables; indexes are used to optimize the speed of data retrieval operations (queries).
 - MySQL – a storage engine for database management.
 - Write-Ahead Logging (WAL) – used to ensure data integrity and recovery.
- Introduce **data persistence** and discuss its mechanisms.
 - Flat Files – must be manually managed for consistency and integrity.
 - Binary Files – compact and efficient for specific data types.
 - Relational Databases – data stored in tables within a database, managed by a DBMS.
 - NoSQL Databases – data stored in various formats, designed for specific use cases, often unstructured data.
 - ACID Properties – (Atomicity, Consistency, Isolation, Durability) ensures transactions are reliable and consistent even during a system failure.
 - Backups – should be periodic, to stay prepared in the event of data loss or corruption.
 - Recovery Procedures – some systems have mechanisms to restore data from backups in the event of a failure.

- Database Replication – data copied from one server to another to ensure availability and disaster recovery.

Data Availability

- Introduce **data availability**. Discuss cloud storage vs. local storage and online access vs. offline access. Highlight advantages and disadvantages of each.
 - Students should recognize both storage and access are individual capabilities. Storage is where or how the data is stored. Access is how the user retrieves and interacts with the data.

Activity: Storage and Access (20-25 minutes)

- In this activity, students will read scenarios and determine which storage method or which access method would best fit the situation and explain why.
- Display Activity Slides 5.2.1 - Storage and Access; students will record their answers on the Student Handout 5.2.1 – Database Fundamentals.
- Decide how you will cover the scenarios; you can read them aloud, call on students to read them aloud, or have students read them independently. Provide students with time to write their answers.
- Students may have different ideas for why they chose their answer.
- Scenarios and Answers:

- A marketing agency with team members spread across multiple countries needs to work on the same project files. Will the marketing agency need cloud storage or local storage? Why?

The marketing agency will need cloud storage. Cloud storage will allow team members to access files from anywhere with an internet connection and receive real-time updates and version history when collaborating on a project.

- A government agency working with classified or sensitive national security data needs to store files that, by regulation, cannot be transmitted over the internet or stored on third-party servers. Will the government agency need cloud storage or local storage? Why?

The government agency will need local storage. Local storage will give the agency full control over access to their devices and reduce the risk of data exposure through online breaches or cloud service vulnerabilities.

- A photographer is working in a remote location with no reliable internet connection and needs to store and back up large RAW image files immediately. Will the photographer need cloud storage or local storage? Why?

The photographer will need local storage. Local storage will allow the photographer to access their files quickly without relying on an internet connection and avoid ongoing subscription fees that come with cloud storage.

- A small business stores customer data, financial records, and inventory logs locally on office computers and external hard drives. A sudden hardware failure, fire, or theft could result in permanent data loss. Will the small business need cloud storage or local storage? Why?

The small business will need cloud storage. Cloud storage will allow the small business to schedule automated backups, scale up or down based on their needs, and avoid the possibility of devices being lost, stolen, or damaged onsite.

- A group of friends wants to play an online multiplayer game where players from different parts of the world join a shared server. Will the group of friends need online access or offline access? Why?

The group of friends will need online access. Online access will allow players to participate in multiplayer games through a shared server and receive live updates during game play.

- A company is hosting a live video conference with remote employees and clients through platforms like Zoom, Microsoft Teams, or Google Meet. Will the company need online access or offline access? Why?

The company will need online access. Online access will allow participants to join a video conference from different locations with an internet connection and communicate in real-time using video, audio, and screen-sharing features.

- A traveler is exploring a new city overseas where mobile data is expensive or unavailable, and Wi-Fi is spotty. Will the traveler need online access or offline access? Why?

The traveler will need offline access. Offline access will allow the traveler to access data without an internet connection and avoid expensive charges. The traveler will need to download offline maps and any other offline services to prepare for their trip.

- A person is on a long international flight with no reliable Wi-Fi or cell signal. Will the person need online access or offline access? Why?

The person will need offline access. Offline access will allow the person to access downloaded content, such as music or a podcast, on their device without relying on an internet connection.

Lab: 5.2.1 SQL Databases (30-45 minutes)

- This lab requires access to the CYBER.ORG Range, the Ubuntu Linux Machine, and Lab Slides 5.2.1 SQL Databases.
- The lab can be performed as a class activity or an independent student assignment.
- Introduce the guiding question and objectives for the lab.
- Students will use the Ubuntu Linux Machine to interact with database files using SQLite3. During the lab, students will create a database from an existing csv file, import the csv file, change the settings of the database, run multiple queries to manipulate the data, and export new csv files with changes made.
- Discuss SQL Key Terms, previously mentioned throughout the lesson.
- In the Ubuntu VM, start by opening the terminal window. Students should navigate to the directory containing the csv file that will be used to create the database. After the prompt changes to `sqlite>`, the terminal will still create and export files to the Database_Fundamentals-lab folder.
- Students should complete the lab portion of their student handout.

Helpful Tips

- Remind students to take their time when typing in the terminal during this lab. Missing even one letter can cause them to retype an entire command.
- If **.mode csv** is not used before importing the file, it will not import correctly, and students will not be able to format the data to columns.
- Students should put a semicolon (;) at the end of each query before hitting the ENTER button, or the query will not run.
- Using the asterisk (*) when running a SELECT query, will select all the data from the corresponding table.
- If students want to cycle through a previous command they can hit the up arrow on the keyboard. This will be useful when using repeat commands such as **SELECT * FROM employees;**
- If the **BEGIN TRANSACTION**; command is not used before making changes to the data, students will not be able to “undo” a mistake or save a change.
 - **ROLLBACK**; can only be used if changes were not saved using the **COMMIT**; or an error has not occurred. **ROLLBACK**; must be used directly after a mistake is made.
 - Students should use **COMMIT**; to save any important changes that are correctly made. If they use **ROLLBACK**; without saving it will undo any changes made since the transaction began.
 - Using **COMMIT**; or **ROLLBACK**; will end the tracking of a transaction. Students must use **BEGIN TRANSACTION**; to start tracking changes again.
- When exporting a file, the mode will determine the format of the file.
- The output command will create the file inside the folder, but the data will not be extracted from the database until the SELECT query is run.
- The SELECT query determines what and how the data is extracted from the database and into the file.

Putting It All Together

- Why is it important to use the BEGIN command before making changes in the database?
Using the BEGIN TRANSACTION command will track the changes being made in the database.
- What command can be used to "undo" changes being tracked with the BEGIN command?
The ROLLBACK command can be used to undo any changes that are tracked with the BEGIN TRANSACTION command, if it has not been saved or an error has not occurred.
- What is the importance of the FROM clause when running a SELECT query?
The FROM clause when running a SELECT query is important because it tells the database what table to retrieve the data from.
- What is the importance of the WHERE clause when running an UPDATE query?
The WHERE clause when running an UPDATE query is important because it locates which specific record in the database is being updated.

Putting It All Together (5 - 10 minutes)

- Review what students learned today by discussing the following questions as a class.
- Why are databases useful?

Databases are useful for gathering, storing, and retrieving data quickly and efficiently. Databases also provide a logical structure to data. When using a database a user can run queries to easily find the information they are looking for and work with multiple tables at once.

- What are some advantages of using a database over a flat file?

Databases are a better solution for anyone handling large volumes of data because they allow for multiple concurrent users, high scalability, quick responses, and support for a variety of data types.

- What is the difference between a record and data values?

A record is a collection of fields that contain data about a given entity while data values are the entries in each field that create a record.

- What should someone consider when choosing between cloud storage and local storage?

When choosing a storage method, a user should consider if they will have access to an internet connection, who will be accessing the data and their location(s), the sensitivity of the data, and any maintenance requirements such as backups and updates.